

Fall 2004, CIS, Temple University

CIS527: Data Warehousing, Filtering, and Mining

Lecture 3

- Data Warehousing and OLAP Technology for Data Mining

Lecture slides taken/modified from:

– Vipin Kumar (<http://www-users.cs.umn.edu/~kumar/csci5980/index.html>)

What is Data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
 - Object is also known as record, point, case, sample, entity, or instance

Objects

Attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute Values

- Attribute values are numbers or symbols assigned to an attribute
- Distinction between attributes and attribute values
 - Same attribute can be mapped to different attribute values
 - Example: height can be measured in feet or meters
 - Different attributes can be mapped to the same set of values
 - Example: Attribute values for ID and age are integers
 - But properties of attribute values can be different
 - ID has no limit but age has a maximum and minimum value

Types of Attributes

- There are different types of attributes
 - **Nominal**
 - Examples: ID numbers, eye color, zip codes
 - **Ordinal**
 - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
 - **Interval**
 - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
 - **Ratio**
 - Examples: temperature in Kelvin, length, time, counts

Properties of Attribute Values

- The type of an attribute depends on which of the following properties it possesses:
 - Distinctness: $= \neq$
 - Order: $< >$
 - Addition: $+ -$
 - Multiplication: $* /$
 - Nominal attribute: distinctness
 - Ordinal attribute: distinctness & order
 - Interval attribute: distinctness, order & addition
 - Ratio attribute: all 4 properties

Attribute Type	Description	Examples	Operations
Nominal	The values of a nominal attribute are just different names, i.e., nominal attributes provide only enough information to distinguish one object from another. ($=$, $\neq$)	zip codes, employee ID numbers, eye color, sex: { <i>male</i> , <i>female</i> }	mode, entropy, contingency correlation, χ^2 test
Ordinal	The values of an ordinal attribute provide enough information to order objects. ($<$, $>$)	hardness of minerals, { <i>good</i> , <i>better</i> , <i>best</i> }, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. ($+$, $-$)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
Ratio	For ratio variables, both differences and ratios are meaningful. ($*$, $/$)	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	geometric mean, harmonic mean, percent variation

Attribute Level	Transformation	Comments
Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function.	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by {0.5, 1, 10}.
Interval	$new_value = a * old_value + b$ where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

Discrete and Continuous Attributes

- Discrete Attribute
 - Has only a finite or countably infinite set of values
 - Examples: zip codes, counts, or the set of words in a collection of documents
 - Often represented as integer variables.
 - Note: binary attributes are a special case of discrete attributes
- Continuous Attribute
 - Has real numbers as attribute values
 - Examples: temperature, height, or weight.
 - Practically, real values can only be measured and represented using a finite number of digits.
 - Continuous attributes are typically represented as floating-point variables.

Types of Data Sets

- **Record**
 - Data Matrix
 - Document Data
 - Transaction Data
- **Multi-Relational**
 - Star or snowflake schema
- **Graph**
 - World Wide Web
 - Molecular Structures
- **Ordered**
 - Spatial Data
 - Temporal Data
 - Sequential Data

Important Characteristics of Structured Data

– Dimensionality

- Number of attributes each object is described with
- Challenge: high dimensionality (curse of dimensionality)

– Sparsity

- Sparse data: values of most attributes are zero
- Challenge: sparse data call for special handling

– Resolution

- Data properties often could be measured with different resolutions
- Challenge: decide on the most appropriate resolution (e.g. “Can’t See the Forest for the Trees”)

Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

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Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document Data

- Each document becomes a ‘term’ vector,
 - each term is a component (attribute) of the vector,
 - the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

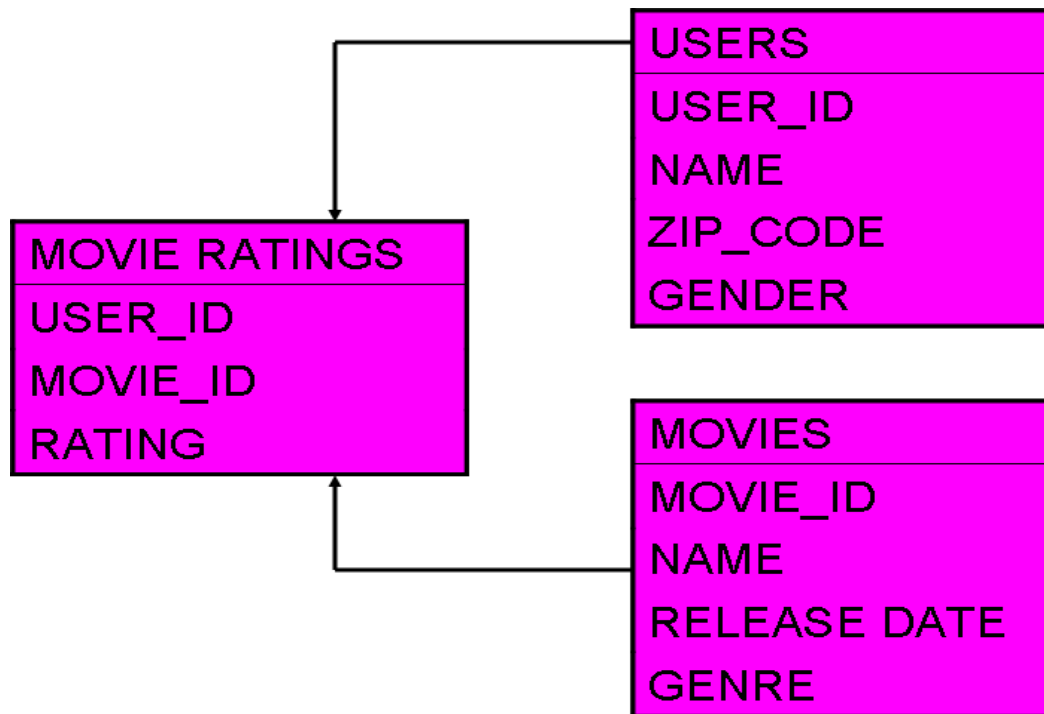
Transaction Data

- A special type of record data, where
 - each record (transaction) involves a set of items.
 - E.g., consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

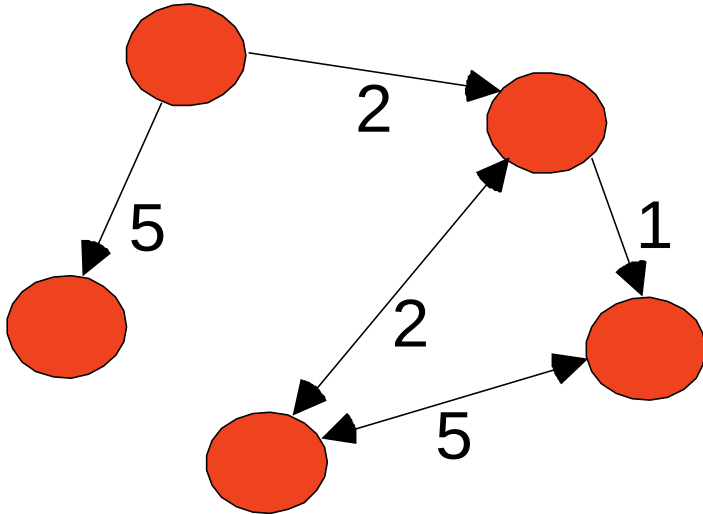
Multi-Relational Data

- Attributes are objects themselves



Graph Data

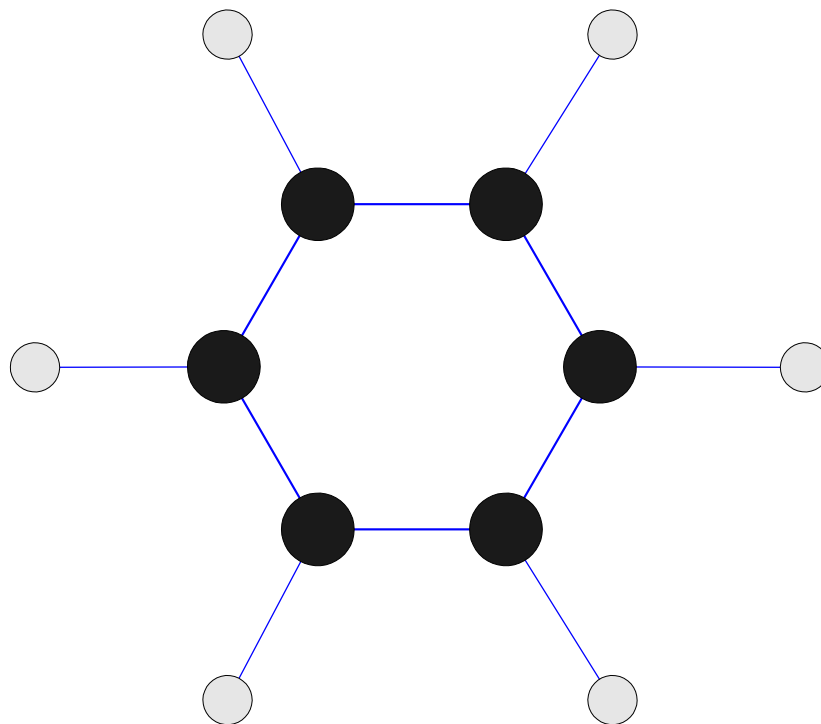
- Examples: Generic graph and HTML Links



```
<a href="papers/papers.html#bbbb">  
Data Mining </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Graph Partitioning </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Parallel Solution of Sparse Linear System of Equations </a>  
<li>  
<a href="papers/papers.html#ffff">  
N-Body Computation and Dense Linear System Solvers
```


Chemical Data

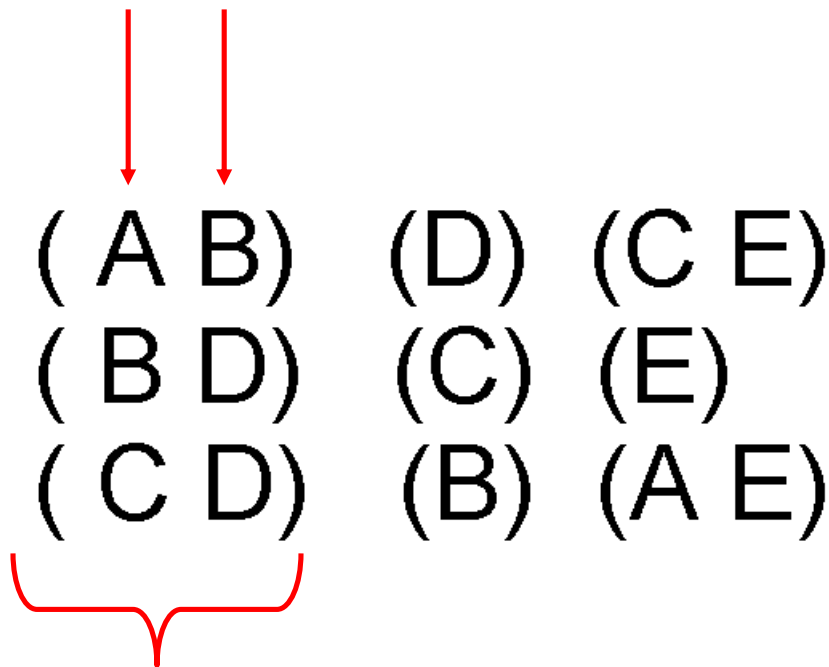
- Benzene Molecule: C_6H_6



Ordered Data

- Sequences of transactions

Items/Events



An element of
the sequence

Ordered Data

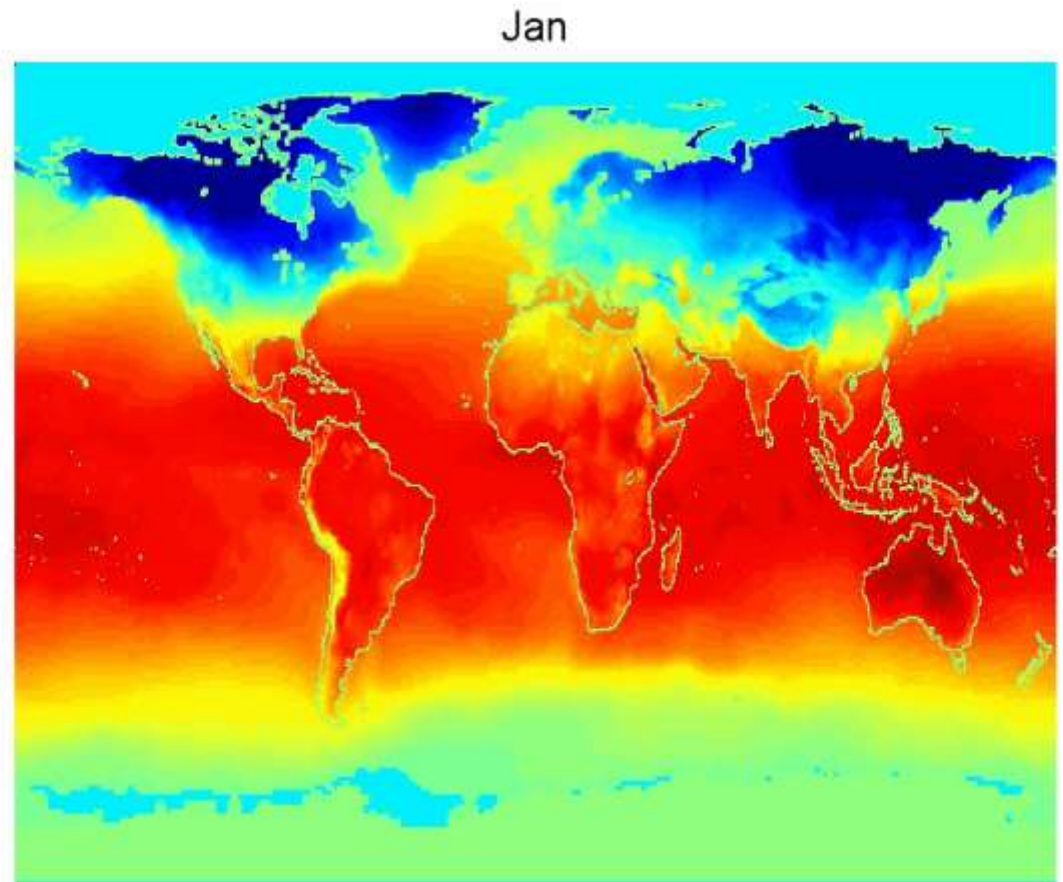
- Genomic sequence data

**GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG**

Ordered Data

- Spatial-Temporal Data

**Average Monthly
Temperature of
land and ocean**

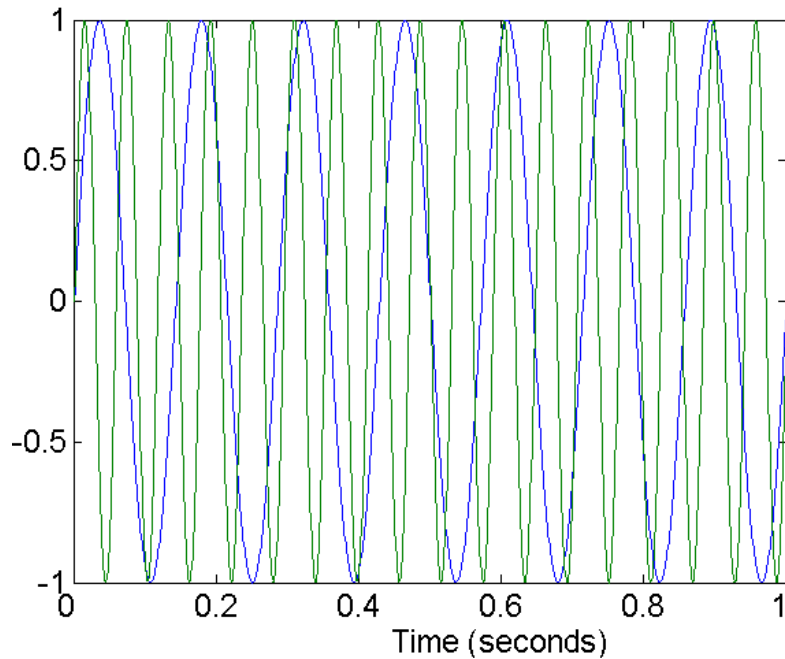


Data Quality

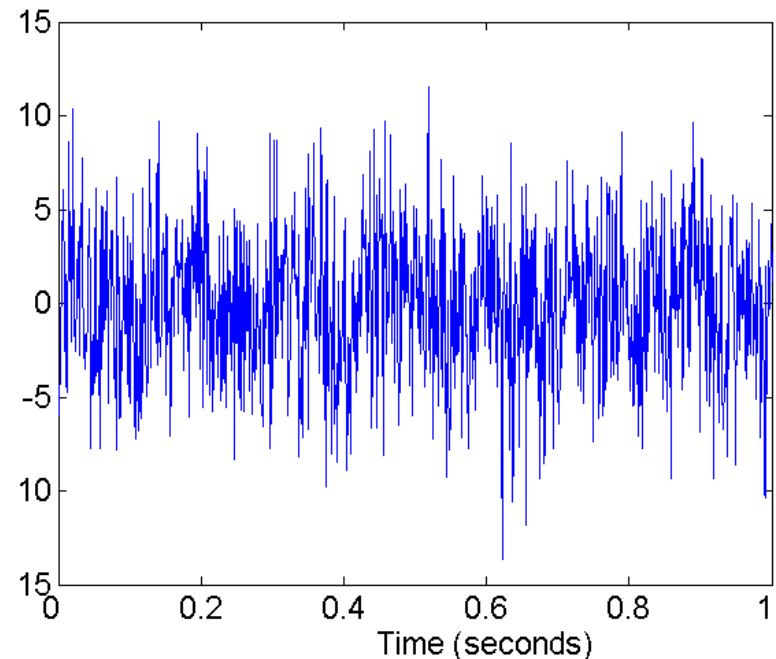
- What kinds of data quality problems?
- How can we detect problems with the data?
- What can we do about these problems?
- Examples of data quality problems:
 - Noise and outliers
 - missing values
 - duplicate data

Noise

- Noise refers to modification of original values
 - Examples: distortion of a person's voice when talking



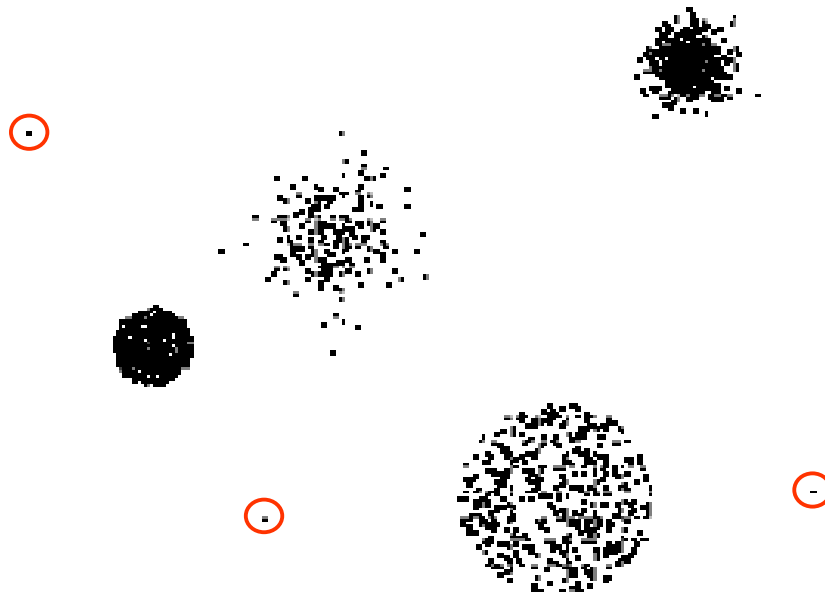
Two Sine Waves



Two Sine Waves + Noise

Outliers

- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set



Missing Values

- Reasons for missing values
 - Information is not collected
(e.g., people decline to give their age and weight)
 - Attributes may not be applicable to all cases
(e.g., annual income is not applicable to children)
- Handling missing values
 - Eliminate Data Objects
 - Estimate Missing Values
 - Ignore the Missing Value During Analysis

Duplicate Data

- Data set may include data objects that are duplicates, or almost duplicates of one another
 - Major issue when merging data from heterogeneous sources
- Examples:
 - Same person with multiple email addresses
- Data cleaning
 - Process of dealing with duplicate data issues

Data Preprocessing

- Aggregation
- Sampling
- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

Aggregation

- Combining two or more attributes (or objects) into a single attribute (or object)
- Purpose
 - Data reduction
 - Reduce the number of attributes or objects
 - Change of scale
 - Cities aggregated into regions, states, countries, etc
 - More “stable” data
 - Aggregated data tends to have less variability

Sampling

- Sampling is the main technique employed for data selection.
 - It is often used for both the preliminary investigation of the data and the final data analysis.
- Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming.
- Sampling is used in data mining because **processing** the entire set of data of interest is too expensive or time consuming.

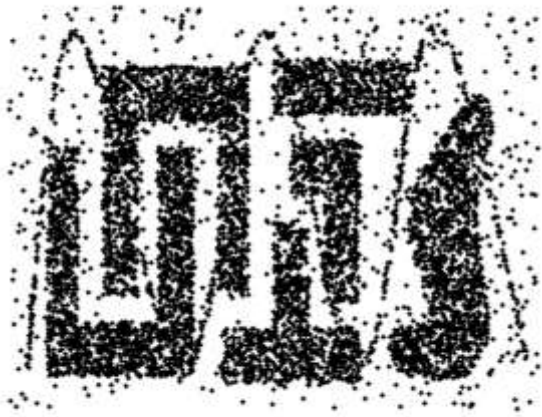
Sampling ...

- The key principle for effective sampling is the following:
 - using a sample will work almost as well as using the entire data sets, if the sample is representative
 - A sample is representative if it has approximately the same property (of interest) as the original set of data

Types of Sampling

- Simple Random Sampling
 - There is an equal probability of selecting any particular item
- Sampling without replacement
 - As each item is selected, it is removed from the population
- Sampling with replacement
 - Objects are not removed from the population as they are selected for the sample.
 - In sampling with replacement, the same object can be picked up more than once
- Stratified sampling
 - Split the data into several partitions; then draw random samples from each partition

Sample Size



8000 points



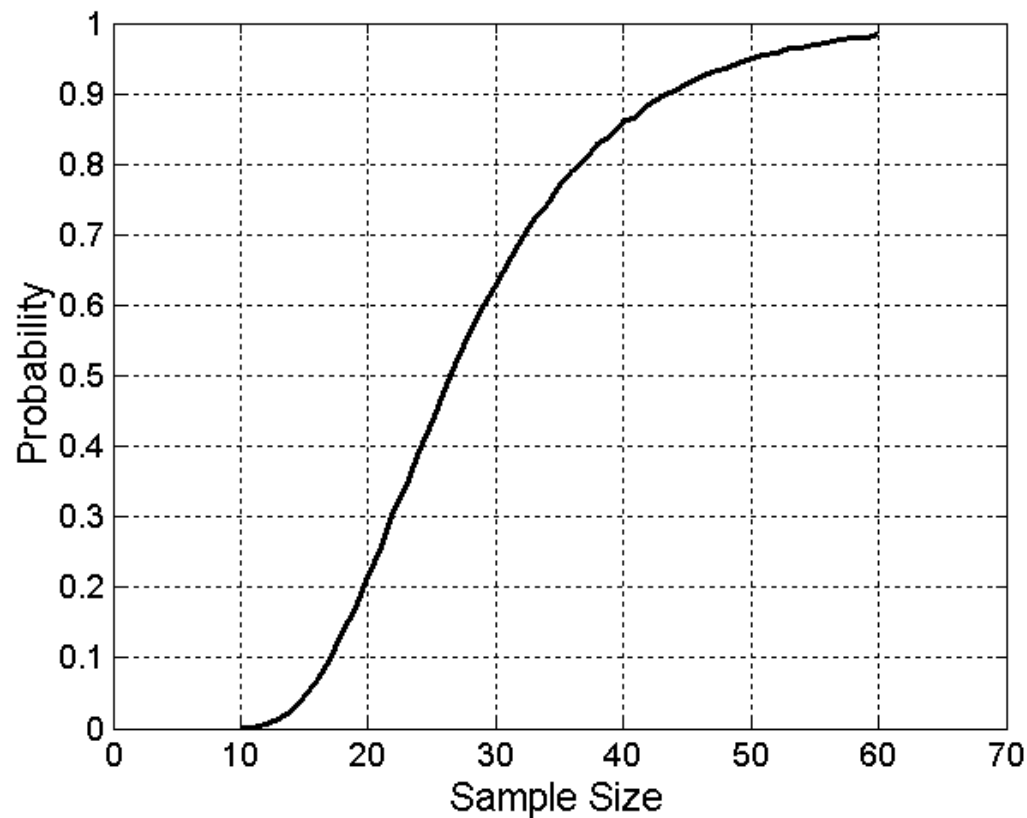
2000 Points



500 Points

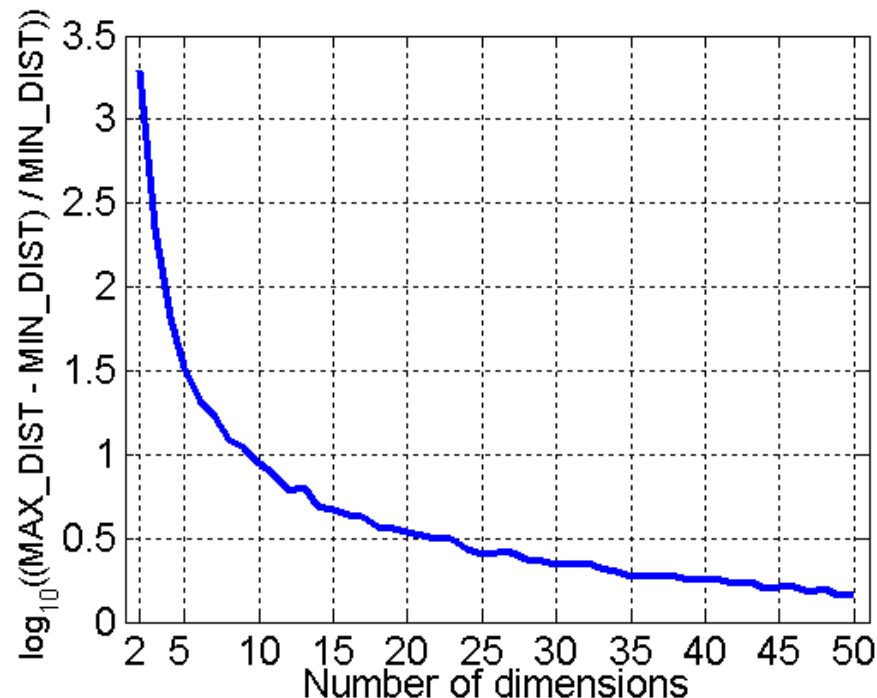
Sample Size

- **What sample size is necessary to get at least one object from each of 10 groups.**



Curse of Dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies
- Definitions of density and distance between points, which is critical for clustering and outlier detection, become less meaningful



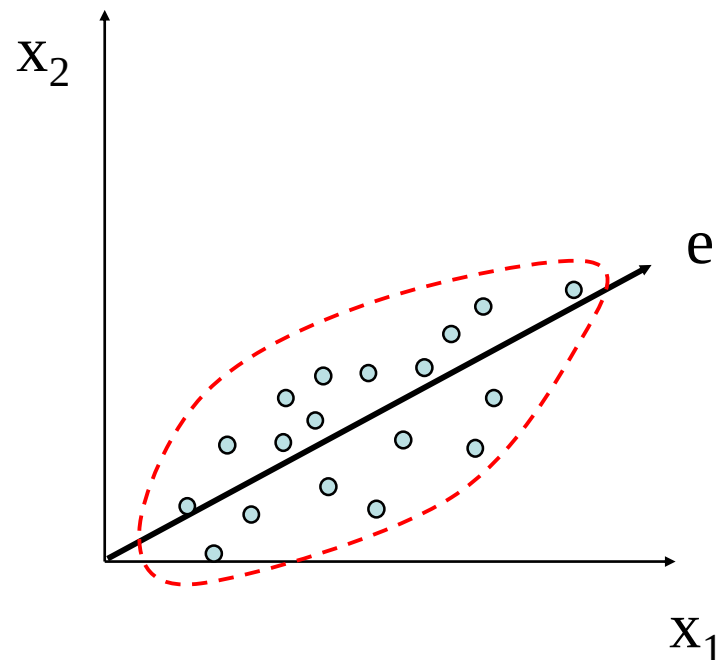
- Randomly generate 500 points
- Compute difference between max and min distance between any pair of points

Dimensionality Reduction

- Purpose:
 - Avoid curse of dimensionality
 - Reduce amount of time and memory required by data mining algorithms
 - Allow data to be more easily visualized
 - May help to eliminate irrelevant features or reduce noise
- Techniques
 - Principle Component Analysis
 - Singular Value Decomposition
 - Others: supervised and non-linear techniques

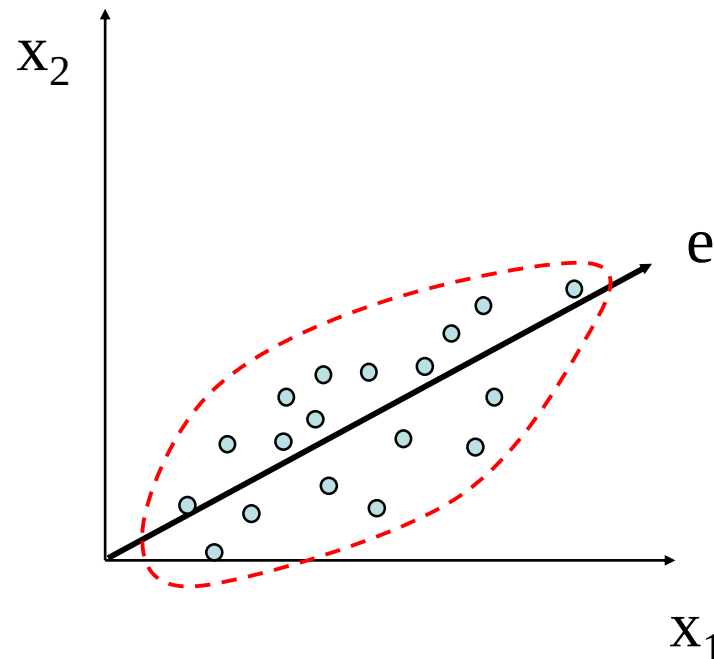
Dimensionality Reduction: PCA

- Goal is to find a projection that captures the largest amount of variation in data



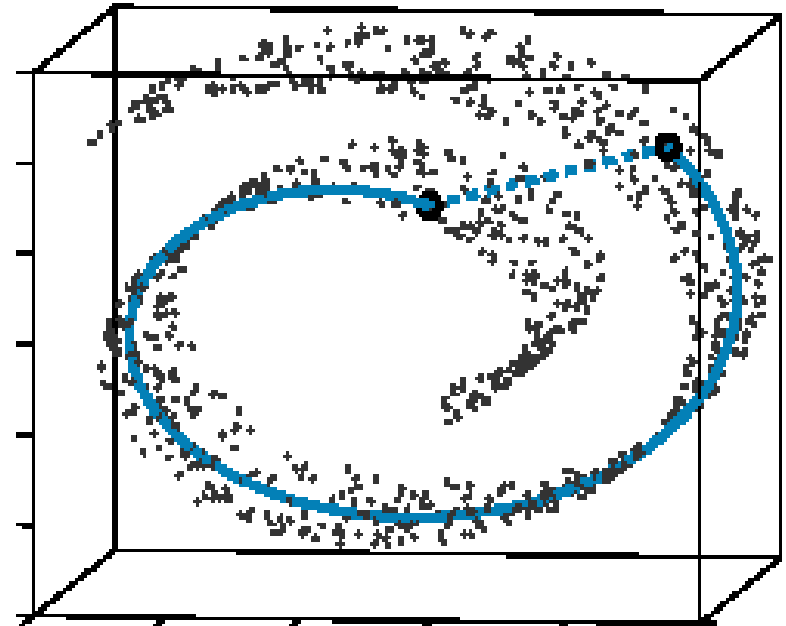
Dimensionality Reduction: PCA

- Find the eigenvectors of the covariance matrix
- The eigenvectors define the new space



Dimensionality Reduction: ISOMAP

By: Tenenbaum, de Silva,
Langford (2000)



- Construct a neighbourhood graph
- For each pair of points in the graph, compute the shortest path distances – geodesic distances

Feature Subset Selection

- Another way to reduce dimensionality of data
- Redundant features
 - duplicate much or all of the information contained in one or more other attributes
 - Example: purchase price of a product and the amount of sales tax paid
- Irrelevant features
 - contain no information that is useful for the data mining task at hand
 - Example: students' ID is often irrelevant to the task of predicting students' GPA

Feature Subset Selection

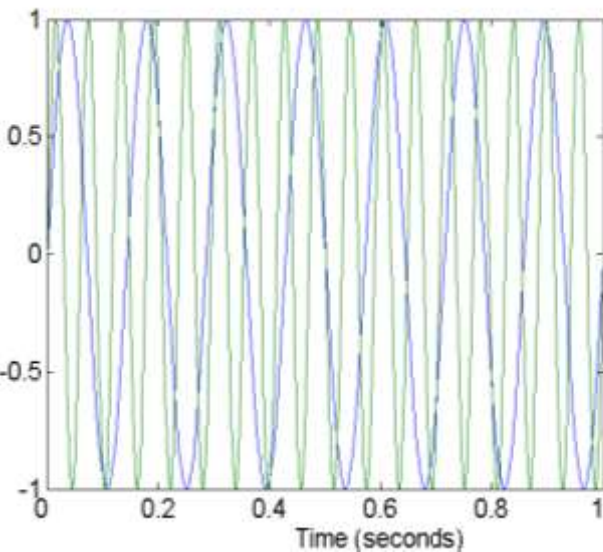
- Techniques:
 - Brute-force approach:
 - Try all possible feature subsets as input to data mining algorithm
 - Embedded approaches:
 - Feature selection occurs naturally as part of the data mining algorithm
 - Filter approaches:
 - Features are selected before data mining algorithm is run
 - Wrapper approaches:
 - Use the data mining algorithm as a black box to find best subset of attributes

Feature Creation

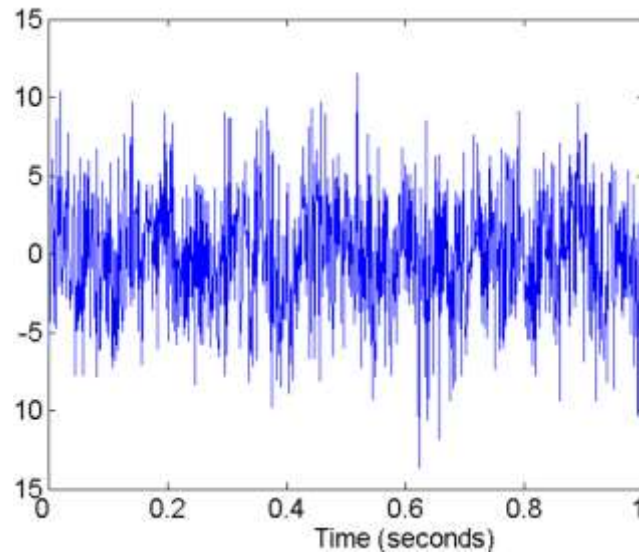
- Create new attributes that can capture the important information in a data set much more efficiently than the original attributes
- Three general methodologies:
 - Feature Extraction
 - domain-specific
 - Mapping Data to New Space
 - Feature Construction
 - combining features

Example: Mapping Data to a New Space

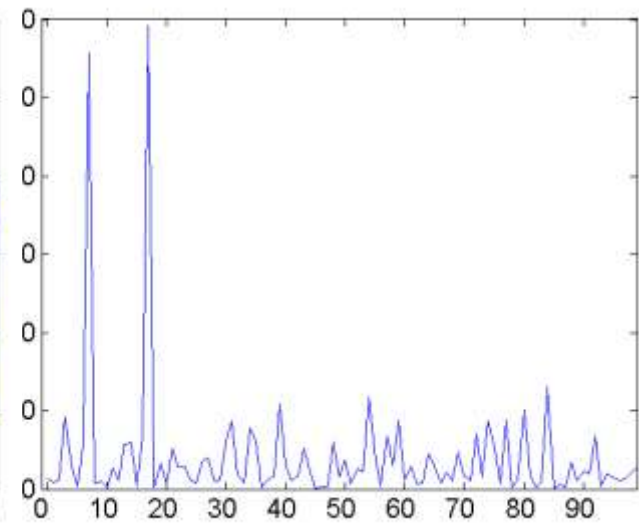
- **Fourier transform**
- **Wavelet transform**



Two Sine Waves



Two Sine Waves + Noise



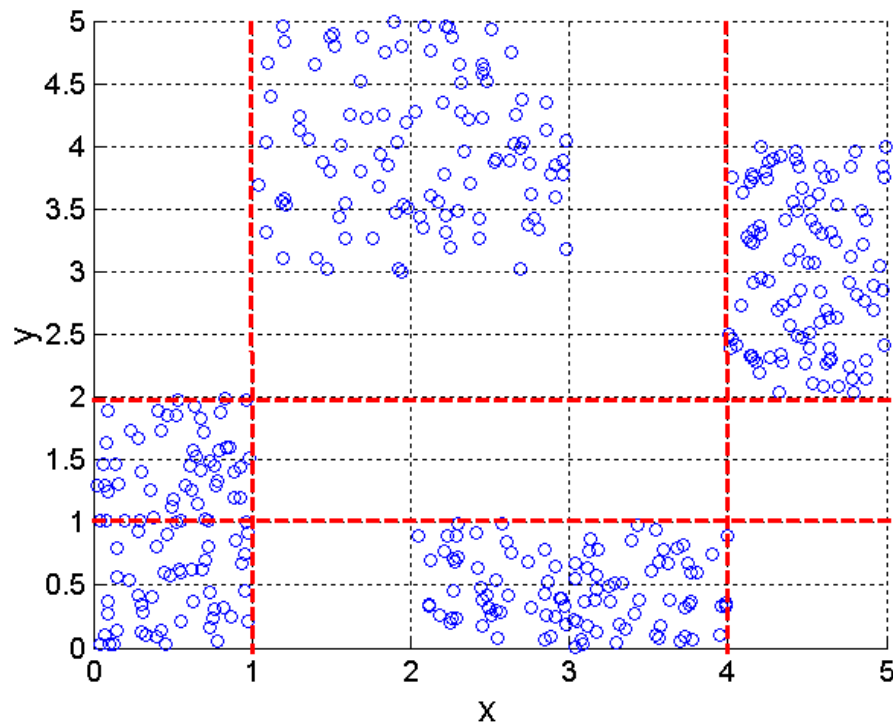
Frequency

Discretization and Binarization

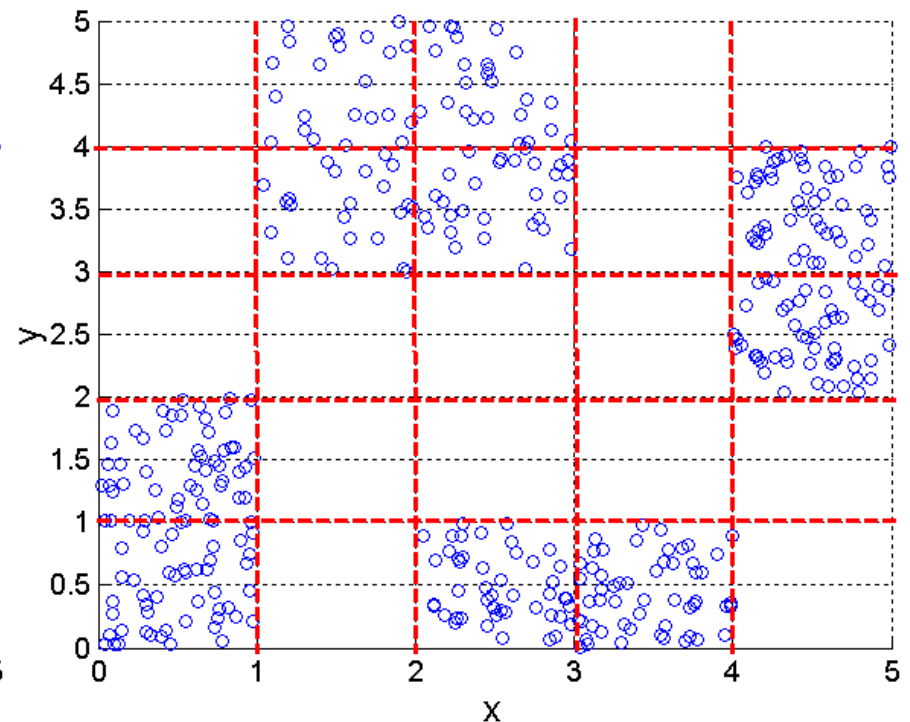
- Different data mining applications require specific data formats
 - Categorical only (discretization)
 - Binary only (binarization)
 - Interval/Ratio only (binarization)
- Discretization: transforming interval attribute into categorical
- Binarization: transforming non-binary attribute into a set of binary attributes

Discretization Using Class Labels

- **Entropy based approach**



3 categories for both x and y



5 categories for both x and y

Attribute Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values
 - Simple functions: x^k , $\log(x)$, e^x , $|x|$
 - Standardization and Normalization

